

Biological random graph models

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PEZ Graduate Program

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Agenda

Biological
random graph
models

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Graph mining

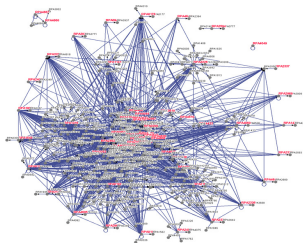
Characteristics

Distributions

Models

Topics

- Graph mining
- Characteristics
- Distributions
- Models



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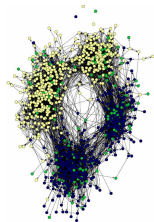
Characteristics

Distributions

Models

Topics

- Graph mining
- Characteristics
- Distributions
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Overview

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Graph mining

Characteristics

Distributions

Models

Applications

- Distributed processing.
- Social media.
- Spatial databases.
- Protein networks.

Queries

- | | |
|----------------------|-------------------------|
| • Neighborhood size. | • Connected components. |
| • Shortest paths. | • Partitioning. |
| • Minimum cuts. | • Bipartiteness. |
| • Maximum flow. | • Circles. |

Challenges

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Computational

- Memory requirements.
- Data is distributed across:
 - Memory hierarchy.
 - Multiple disks.
 - Network.

Algorithmic

- Reorganization.
- Randomization.
- Heuristics.
- Visualization.

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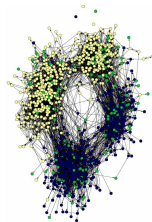
Characteristics

Distributions

Models

Topics

- Graph mining
- **Characteristics**
- Distributions
- Models



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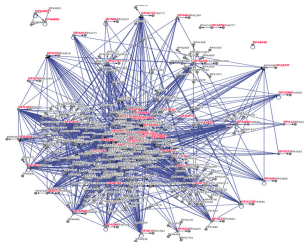
Characteristics

Distributions

Models

Characteristics

- Self similar and scale free.
- Shrinking diameter.
- Giant component.
 - Communities.
- Densification.
- Preferential attachment.
- Heavy tails in degree distributions.



Scale free graphs

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Distributions

Models

Definition

Let $g(n)$ be the graph growth function. Then

$$g(\beta_0 n) = \tilde{g}(\beta_0) g(n)$$

$$\Leftrightarrow$$

$$\lim_{n \rightarrow +\infty} \left(\frac{g(\beta_0 n)}{g(n)} \right) = \Theta(1)$$

Notes

- The most fundamental property.
- Allows self-similar graphs.
- Contributes to overall graph robustness.

Shrinking diameter

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Distributions

Models

Definition

- Maximum shortest path length decreases.

Notes

- Prone to outliers.
- Effective diameter.
- Small world phenomena.
- Robustness.



Densification

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Characteristics

Distributions

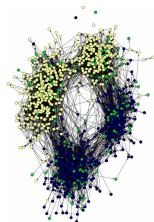
Models

Definition

- Ratio of edges to vertices is an increasing time function.

Notes

- Average degree $\langle d \rangle$ increases.
- $\langle d \rangle$ exact behaviour is of interest.
- Degree distribution is of interest too.



Giant connected component

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Characteristics

Distributions

Models

Definition

Graph connectivity threshold point t_0 :

- Many, medium components before t_0 .
- One, large component after t_0 .

Notes

- t_0 depends on average degree $\langle d \rangle$.
- t_0 is a phase transition point.
- Phase transition in connectivity pattern.
- Interesting patterns around t_0 .
- Similar to phase change from complexity theory.

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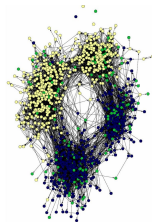
Characteristics

Distributions

Models

Topics

- Graph mining
- Characteristics
- **Distributions**
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Distributions

Models

Significant graph metric

- Defines graph topology.
- Affects graph growth rate.
- Affects self-similarity.

Notes

- Degree distribution alone cannot describe growth pattern.
- Vertex degree-degree correlation is more informative.
- Vertex degree ranking is of interest.
- No completely satisfactory model exists.

Exponential

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Characteristics

Distributions

Models

Definition

$$\text{prob} \{X = k\} = \lambda_0 e^{-\lambda_0 k} \quad \lambda_0 > 0$$

Notes

- Tractable bounds.
- Closed forms.
- Memoryless property.
- Quick decay.
- Last two properties undesired for self similarity.

Binomial

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Characteristics

Distributions

Models

Definition

$$\text{prob} \{X = k\} = \binom{n}{k} p_0^k (1 - p_0)^{1-k} \quad 0 < p_0 < 1$$

Notes

- n is the number of vertices.
- Some early models lead to it.
- When n grows large, it is approximated by exponential.
- Quick decay.

Poisson

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Characteristics

Distributions

Models

Definition

$$\text{prob} \{X = k\} = e^{-\langle d \rangle} \frac{\langle d \rangle^k}{k!}$$

Notes

- Other early models lead to it.
- Also approximated by exponential.
- Also quick decay.

Power law

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Characteristics

Distributions

Models

Definition

$$\text{prob} \{X = k\} = P(k) = \alpha_0 k^{-\gamma_0} \quad \alpha_0 > 0, \gamma_0 \geq 1$$

Notes

- α_0 is a normalization constant.
- γ_0 is termed the system exponent.
 - Characterizes graph subclasses.
 - Connections to Lyapunov exponent.
- Zipf law for $\gamma_0 = 1$.
 - Information retrieval.
- Lotka law for $\gamma_0 = 2$.
 - Scientific citation networks.

Power law

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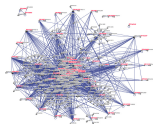
Characteristics

Distributions

Models

Notes

- When $\gamma_0 < 2$, then the average degree diverges.
- When $\gamma_0 < 3$, then the standard deviation diverges.
- Empirically for most scale free graphs $2 < \gamma_0 < 3$.
- Real world graph properties have been attributed to divergence.
 - Forces graph to be finite.



Cutoff power law

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Distributions

Models

Definition

$$\text{prob} \{X = k\} = \alpha_0 (k + k_0)^{-\gamma_0} e^{-\frac{k}{k_1}} \quad \alpha_0 > 0, \gamma_0 \geq 1$$

Notes

- α_0 and γ_0 as before.
- k_0 and k_1 need to be determined.
- k_0 is the degree threshold.
- Exponential decay through k_1 only for large degrees.
- Fits best to actual data.
- Upper and lower degree bounds are important.

Multifractal

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Distributions

Models

Definition

$$\text{prob} \{X = k\} = \sum_{i=1}^p w_i P_i(k), \quad \sum_{i=1}^p w_i = 1$$

Notes

- Weighted power law sum.
- More general than power law.
- Harder to fit.
- Recent model.
- Promising early results.

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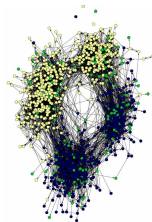
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Distributions

Models

Topics

- Graph mining
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Complementary approaches

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Distributions

Models

Power distributions are given

- Locate large graphs that possess power distributions.
- Derive properties concerning deeper structure.
 - Black box approach.
 - Closer to real world problems.
 - Domain specific.

Power distributions are generated

- Generate small graphs that possess power distributions.
- Find evolution model ensuring power distributions.
- Monitor graph properties and structure.
 - White box approach.
 - Generic and flexible.
 - More demanding.
 - Overmodeling and undermodeling issues.

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Distributions

Models

Random models

- $\mathcal{G}_{n,p}$ or Erdős - Rényi.
- Albert - Barabási.
- Gene duplication.
- Kronecker.
 - Deterministic.
 - Probabilistic.
- Aiello family.

Other models

- $\tilde{\mathcal{G}}^n$.
- $\mathcal{G}_{n,m}$.
- Kumar family.
- R-MAT.
- Jelly point.

$\mathcal{G}_{n,p}$ model

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Distributions

Models

Overview

- Static.
- Probabilistic.
- Each of the $\binom{n}{2}$ edge exists independently with probability p .

Notes

- Easy interpretation.
- Easy generation.
- Closed forms for most characteristics.
- Independent edges lead to exponential distribution.
- No preferential attachment.

Albert - Barabási model

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Characteristics

Distributions

Models

Overview

- Dynamic.
- Probabilistic.
- Number of vertices and edges evolves.
 - Start with n_0 connected vertices
 - Add a new vertex of random degree $\tau < n_0$.
 - Select τ vertices with probability proportional to their degree.

Notes

- Easy interpretation.
- Preferential attachment.
- Uniform selection probability leads to exponential distribution.

Gene duplication model

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Distributions

Models

Overview

- Start with a graph of n_0 genes.
- Select uniformly a gene g_0 .
- Duplicate g_0 .
- With probability p_0 remove an edge from g_0 .
 - Edges are uniformly selected.
- With probability q_0 add an edge to g_0 .
 - Genes are uniformly selected.

Notes

- Dynamic.
- Probabilistic.
- Preferential attachment.
- Leads to power law.

Kronecker model

Deterministic variant

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Graph mining

Characteristics

Distributions

Models

Overview

- Graphs are represented by adjacency matrices.
- Kronecker products are heavily used.

Generation

- Select a generator graph.
 - Not an easy topic.
- Leads to self similar graphs.
- Has most known properties.
- Generalizes R-MAT model.

Kronecker model

Deterministic variant

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Distributions

Models

Adjacency matrix

$$\mathbf{A}[i, j] \triangleq \begin{cases} 1, & (v_i, v_j) \in E \\ 0, & (v_i, v_j) \notin E \end{cases}$$

Linear-algebraic aspect

- Time evolving.
- Adjacency matrix density.
- Power distribution in:
 - Adjacency matrix eigenvalues.
 - Principal eigenvector components.
- Alternative representations:
 - Incidence matrix.
 - Circuit matrix.
 - (Normalized) Laplace matrix.
 - König matrix.

Kronecker model

Analysis

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Kronecker product

For $\mathbf{A} \in \mathbb{R}^{m_1 \times n_1}$ and $\mathbf{B} \in \mathbb{R}^{m_2 \times n_2}$:

$$\mathbf{A} \otimes \mathbf{B} \triangleq \begin{bmatrix} a[1,1] \mathbf{B} & a[1,2] \mathbf{B} & \dots & a[1,n_1] \mathbf{B} \\ a[2,1] \mathbf{B} & a[2,2] \mathbf{B} & \dots & a[2,n_1] \mathbf{B} \\ \vdots & \vdots & \ddots & \vdots \\ a[m_1,1] \mathbf{B} & a[m_1,2] \mathbf{B} & \dots & a[m_1,n_1] \mathbf{B} \end{bmatrix} \in \mathbb{R}^{m_1 m_2 \times n_1 n_2}$$

where $a[i,j]$ denotes the entry of $\mathbf{A}$ at row i and column j .

Kronecker model

Example

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Generator graph

$$\mathbf{A}^{[0]} = \mathbf{A}^{[1]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

Next step

$$\mathbf{A}^{[2]} = \mathbf{A}^{[1]} \otimes \mathbf{A}^{[0]} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

Kronecker model

Probabilistic variant

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Distributions

Models

Overview

- Replace adjacency matrix with probability matrix.
 - Each row sums to one.
 - Nonzero entries.
 - Row stochastic matrix.
 - Doubly stochastic matrix if symmetric.
- Follow deterministic procedure.
- Result is distribution matrix.

Notes

- More general.
- More flexible.
- Numerical challenges.

Aiello model family

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Distributions

Models

Overview

- Reviews existing random graph models.
- Provides a partial framework for building random graphs.
- Introduces four new random graph models termed A, B, C, and D:
 - Models A and B are simple models for directed graphs.
 - Model C encompasses and extends Models A and B.
 - Model D generalizes Model C for undirected graphs.

Model A

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Distributions

Models

Graph evolution

- Start with n_0 connected vertices.
- With probability $(1 - p_0)$:
 - Add a new vertex v_0 .
 - Set in- and out-degree of v_0 are one.
 - Select one vertex v_{out} and form an edge (v_{out}, v_0) .
 - Increase out-degree of v_{out} by one.
 - Select one vertex v_{in} and form an edge (v_0, v_{in}) .
 - Increase in-degree of v_{in} by one.
 - Both v_{in} and v_{out} are chosen with probability *proportional to their in- or out-degree*.
- With probability p_0 :
 - Select two vertices v_{in} and v_{out} .
 - Add the edge (v_{in}, v_{out})
 - Both v_{in} and v_{out} are chosen with probability *proportional to their in- or out-degree*.

Model A

Analysis

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Models

Key points

- Power distributions for both in- and out-degrees.
- Simplistic model.
- The same exponent for in- and out-degree distributions.

Model B

Overview

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Characteristics

Distributions

Models

Graph evolution

- Almost the same dynamics as model A.
- Use in- and out-weights to select nodes instead of in- and out-degrees.

Model B

Analysis

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Models

Key points

- Power distributions for both in- and out-degrees.
- More flexible model.
- Different exponents for in- and out-degree distributions.

Model C

Overview

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Models

Graph evolution

- Almost the same dynamics as model B.
- When adding a new vertex:
 - $m^{e,e}$ edges randomly.
 - $m^{n,e}$ to the new vertex randomly.
 - $m^{e,n}$ from the new vertex randomly.
 - $m^{n,n}$ loops to the new vertex.

Model C

Analysis

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Distributions

Models

Key points

- Power distributions for both in- and out-degrees.
- Far more flexible model.
- Different exponents for in- and out-degree distributions.

Model D

Overview

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Models

Graph evolution

- Unirected version of model B.
- Still use two weights internally.

Bibliography

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Bibliography



Aiello W., Chung F., and Lu L., *Random evolution in massive graphs*, in Abello J., Pardalos P., and Resende M. editors, *Handbook of massive data sets*, Kluwer, 2002.



Vitter J., *Algorithms and data structures for external memory: Dealing with massive data*, ACM, 2001.



Leskovec J., Chakrabarti D., Kleinberg J., and Faloutsos C., *Realistic, mathematically tractable graph generation and evolution using Kronecker multiplication*, PKDD05, Springer, 2005.

Questions?

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Thank you for your time

